Proper face groups in a Suzuki rotation marking

Human Compression project: a bounded F003 replacement argument

We isolate an obstruction which avoids the intermediate matrix classification in Spiga's supplied preprint [1]. It proves the rank-four exclusion when the field exponent is prime, and identifies precisely the remaining subfield case. It is not a complete recovery of that preprint's all-field theorem. The original theorem and its historical conjecture attribution remain Spiga's; this note grants no new conjecture or whole-original credit.

Put $S = \mathrm{Sz}(q)$, where $q = 2^N$, $N \geq 3$ is odd, and $\theta = 2^{(N+1)/2}$. We retain the following classical Suzuki structure, already used in the project's accepted Suzuki application: the doubly transitive ovoid action; a unique ovoid fixed point for each involution; Sylow 2-groups of exponent four with every involution central and with each involution centralizer a Sylow 2-group; and the maximal-subgroup list consisting of point stabilizers, odd-torus normalizers and proper Suzuki subfield groups. The torus normalizers are $D_{2(q-1)}$ and $C_{q\pm\sqrt{2q}+1} \rtimes C_4$. Their odd kernels are cyclic, and every involution outside that kernel acts on it by inversion. A nonidentity odd-order element has a unique maximal odd torus as its centralizer. These are retained structural premises from Suzuki [2], not classifications reproved here.

In the standard four-dimensional model, the central involutions of the root group at infinity and the Weyl involution are

$$z(s) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ s & 0 & 1 & 0 \\ s^\theta & s & 0 & 1 \end{pmatrix}, \qquad J = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}.$$

The opposite central involutions are $Jz(t)J$. The stabilizer of the ordered pair $(\infty, 0)$ is diagonal, hence has the form $\mathrm{diag}(d, e, e^{-1}, d^{-1})$ with $d, e \neq 0$. The Suzuki relation between d and e will not be needed. These coordinate descriptions are also retained; the calculation below uses only the displayed matrices.

Lemma 1. *The product of two involutions of S is either the identity, an involution, or an element of odd order.*

Proof. The only possible nontrivial even orders in S are two and four, since an even-order element lies in a Sylow 2-group or has an odd-order power which centralizes an involution, and the latter centralizer is a 2-group. If involutions x, y had product of order four, then $\langle x, y \rangle$ would be a dihedral group of order eight. It lies in a Sylow 2-group, where x and y are central and must commute, a contradiction. □

Lemma 2. *Let s, t, d, e be nonzero field elements and put $D = \mathrm{diag}(d, e, e^{-1}, d^{-1})$. Then $z(s)DJz(t)J$ is not an involution.*

Proof. Direct multiplication gives

$$M = z(s)DJz(t)J = \begin{pmatrix} d & 0 & td & t^\theta d \\ 0 & e & 0 & te \\ sd & 0 & std + e^{-1} & st^\theta d \\ s^\theta d & se & s^\theta td & ste + s^\theta t^\theta d + d^{-1} \end{pmatrix}.$$

The $(1, 2)$ entry of M^2 is $st^\theta de$, which is nonzero. The corresponding entry of the identity is zero. □

Theorem 3. *Suppose $S = \langle a, b, c\rangle$ and $u = ab$, $v = bc$, $w = abc$ are involutions. Suppose $|a|, |c| > 2$ and $b^2 \neq 1$. Put $F = \langle a, b\rangle$ and $E = \langle b, c\rangle$. Then F and E cannot both be contained in subgroups which are either point stabilizers or odd-torus normalizers.*

Proof. The relations give

$$a = wv, \qquad c = uw, \qquad F = \langle a, u\rangle, \qquad E = \langle c, v\rangle.$$

By Lemma 1, a, c have odd order. If F lies in an odd-torus normalizer, then a belongs to its odd cyclic kernel and u inverts a. Therefore $b = a^{-1}u$ satisfies $b^2 = 1$, contrary to hypothesis. The same argument with c, v excludes a torus normalizer containing E.

Under the contrary assumption, F and E thus lie in point stabilizers S_P and S_Q. If $P = Q$, the three generators belong to one proper point stabilizer, contrary to generation of S. Conjugate the ordered pair (P, Q) to $(\infty, 0)$. The involutions $u \in F$ and $v \in E$ have these unique fixed points, so $u = z(s)$ and $v = Jz(t)J$ for nonzero s, t. The element $b \in F \cap E$ belongs to the diagonal two-point stabilizer. Consequently $w = ub^{-1}v$ has exactly the form excluded by Lemma 2. This contradiction proves the theorem. □

Corollary 4. *If N is an odd prime, S has no rotation marking of a chiral polytope of rank four.*

Proof. We use the standard rotation presentation and chirality criterion: for a chiral rank-four polytope, a, b, c generate the group, ab, bc, abc are involutions, the two face subgroups F, E above are proper, and there is no group automorphism sending (a, b, c) to (a^{-1}, a^2b, c). The last criterion is the usual extension to the first flag reflection; it is a retained general polytope input. If $a^2 = 1$, the identity automorphism would satisfy it, a contradiction. By duality $c^2 \neq 1$ as well.

If $b^2 = 1$, the relations $(ab)^2 = (bc)^2 = 1$ say that b inverts both a and c. Then $(abc)^2 = 1$ reduces to $ac^{-1}a^{-1}c = 1$, so a and c commute. The group S has the abelian normal subgroup $\langle a, c\rangle$ and quotient of order at most two. It is soluble, contradicting its nonabelian simplicity. Thus all hypotheses of Theorem 3 hold.

When N is prime, the only possible proper Suzuki subfield group is $\mathrm{Sz}(2) \cong C_5 \rtimes C_4$. Its characteristic C_5 has a unique maximal odd torus as centralizer in S; normalizing C_5 therefore normalizes that torus. Hence this subfield group is already contained in an odd-torus normalizer. The retained maximal-subgroup list now puts each proper F, E in one of the two types excluded by Theorem 3. □

For general odd N, the same argument shows that any putative chiral rank-four marking must have at least one of its two face groups inside a proper Suzuki subfield group of parameter at least eight. This is an explicit remaining obstruction, not a proof that it cannot occur. The all-rank, almost-simple reduction used by Spiga retains the earlier Hubard–Leemans theorem; it is not reproved by this note.

The diagnostic programs in the accompanying working directory enumerate the fixed group $\mathrm{Sz}(8)$. They find no generating rotation triple with both face groups proper. They motivated the argument, but are not premises of the written uniform results. No finite search is used to infer the subfield exclusion still required for composite N.

References

[1] P. Spiga, *On the rank of Suzuki polytopes: an answer to Hubard and Leemans*, arXiv:2010.04080v1, 8 October 2020. The fixed source edition has four pages.

[2] M. Suzuki, *On a class of doubly transitive groups*, Annals of Mathematics 75 (1962), 105–145.

[3] I. Hubard and D. Leemans, *Chiral polytopes and Suzuki simple groups*, Rigidity and Symmetry, Fields Institute Communications 70 (2014), 155–175.